

Praveen Kumar Balaji

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PROFESSIONAL SUMMARY

AI Engineer and Data Scientist with 4+ years of experience designing and deploying end-to-end machine learning solutions across public sector, enterprise, and research environments. Skilled in NLP, time series forecasting, anomaly detection, and computer vision, with hands-on expertise in building RAG systems, multi-agent AI architectures, and predictive analytics workflows. Proficient in AWS, Azure, GCP, and MLOps tools, with a strong focus on automation, operational efficiency, and delivering data-driven insights that support strategic decision-making.

PROFESSIONAL EXPERIENCE

NATIONAL ASSOCIATION OF COUNTIES (NACo),

AI Engineer Intern

Washington DC, USA

Oct 2024 – Mar 2024

- Implemented a Retrieval-Augmented Generation (RAG) system using Meta's LLaMA models and AWS services (S3, SageMaker, OpenSearch) to power an internal AI chatbot for NACo, enabling advanced semantic search and automated Q&A across 1,000+ county policy documents—reducing manual research time by 60% and improving information retrieval accuracy by 85%.
- Built and deployed a modular LangChain pipeline for prompt orchestration, document memory, and fallback chaining; integrated Hugging Face and FAISS for embeddings, improving retrieval precision by 30% and reducing latency by 40% across 1,000+ documents.
- Developed an AI-powered research pipeline using Power Automate, Meta's LLaMA, Perplexity API, and OpenAI for web data extraction, validation, and summarization; implemented deduplication and schema-based checks, reducing manual effort by 90% and boosting data accuracy by 40%.
- Engineered a multi-agent, AI-driven architecture to automate anomaly detection, compliance validation, and service ticketing; orchestrated agents like Cyber Compliance Agent and Net Guard to process system logs and network packets, detect violations, and trigger remediation via Ticket Agent and Analytics Agent—reducing incident response time by 70% and increasing operational visibility by 3x.

KYNDRYL,

Data Scientist

Chennai, India

Mar 2022 – Jul 2023

- Designed and deployed machine learning models for product demand forecasting using XGBoost, ARIMA, and Prophet within Azure Machine Learning; integrated time series pipelines with real-time dashboards and applied NLP (topic modeling, sentiment scoring) on distributor messages to reduce stockouts by 22% and increase forecast accuracy by 35%.
- Built interactive BI dashboards in Tableau and Power BI, merging multi-source datasets from MySQL and Oracle Cloud; incorporated ML-driven KPI indicators and anomaly detection for the Henkel Team to drive data-backed decisions across supply chain and procurement.
- Executed end-to-end experimentation frameworks using A/B testing with confidence interval analysis & time series forecasting via Prophet & XGBoost; leveraged text analytics (NLP) to compare hypotheses, improving decision-making and delivering cost savings in supply chain and finance operations.
- Performed advanced analytics & NLP modeling on customer feedback & technical logs, using transformer-based models for sentiment classification and clustering; identified actionable trends that boosted product quality and customer satisfaction, supported by executive-ready reports and visualizations.

JOHNBHAI SOFTCOM PRIVATE LIMITED,

Data Scientist

Chennai, India

Jan 2020- Mar2022

- Applied advanced NLP techniques using Python (NLTK, TextBlob) to analyze over 10,000 customer reviews; implemented Naive Bayes and TF-IDF-based topic modeling to extract sentiment trends & recurring themes, enabling product teams to refine strategy and achieve a 15% drop in customer complaints.
- Built and deployed anomaly detection models using Logistic Regression and Decision Trees with scikit-learn to flag outliers in financial transactions, improving early detection of anomalies and reducing manual audit effort by 30%.
- Designed and orchestrated automated data pipelines via Azure Data Factory, integrating multi-format data from APIs, flat files, and cloud databases into centralized analytics environments—boosting data availability by 40% and enabling real-time insights.
- Developed linear regression-based forecasting solutions to predict monthly revenue using historical sales, campaign performance, and seasonal trends; delivered forecast-driven recommendations that empowered the finance team to optimize budget allocation.
- Led cross-functional business analytics initiatives, partnering with operations, sales, and support teams to define KPIs, build self-service dashboards, and uncover performance gaps—driving 10–15% efficiency gains through data-informed decision-making and continuous monitoring.

EDUCATION

George Mason University, Fairfax- Masters in Data Analytics Engineering (Data Science)

May 2025

Sathyabama University, India - Bachelor of Engineering, Electronics and Communications

May 2020

SKILLS

Python, SQL, Pandas, NumPy, Scikit-learn, NLTK, TextBlob, Prophet, XGBoost, FAISS, Time Series Forecasting, A/B Testing, Sentiment Analysis, Anomaly Detection, Topic Modeling, Retrieval-Augmented Generation (RAG), LLaMA, LangChain, Classification & Regression (Logistic, Linear, XGBoost), Clustering (K-Means, DBSCAN), Deep Learning (CNN, RNN), Transformer models, Reinforcement Learning (basic) Model Evaluation (Cross-validation, ROC-AUC), and Hyperparameter Tuning (Grid Search, Random Search), Azure ML, Azure Data Factory, AWS (S3, SageMaker, OpenSearch), Google Cloud Platform, Docker (for ML model deployment), Hugging Face Transformers, OpenAI API, Perplexity API, Power Automate.

ACADEMIC PROJECTS

Transaction Anomaly Detection using Isolation Forest - Designed and implemented a fraud detection pipeline using the Isolation Forest algorithm and statistical thresholds to flag high-risk financial transactions. Performed exploratory data analysis and feature selection on transaction data, achieving robust anomaly classification. Deployed the trained model, enabling real-time predictions via REST API for production-scale applications.

AI-Driven Market Forecasting for Crypto, Stocks, and Commodities - Built & trained **ARIMA** and **LSTM** models to forecast multi-sector market trends across sectors like crypto, energy, and metals. Utilized **TensorFlow**, **Python**, and **pandas** for time-series modeling & implemented custom feature extraction pipelines. Achieved 89% prediction accuracy, & developed a Streamlit dashboard to visualize forecast outputs for investment decision-making.

Capstone Project - AI-Driven Terrorism Trend Forecasting (GT-TAC, George Mason University) - Collaborating with the Global Terrorism Trends and Analysis Center (GT-TAC) to design an AI-powered system for forecasting global terrorism threats. Developed an end-to-end pipeline using **Python**, **Pandas**, and **Scikit-learn** for data wrangling, feature engineering, and predictive modeling. Applied natural language processing (NLP) techniques and text classification to extract insights from unstructured reports. Built machine learning models for incident severity scoring, spatiotemporal trend analysis, & regional risk classification. Integrated geospatial ML and interactive dashboards to support strategic decision-making in counter-terrorism operations.

PUBLICATIONS

Praveen Kumar Balaji, Karthik R, "Identification and Classification of Plant Disease using Image Processing Techniques," IEEE Publication, May 2020

Designed and implemented a system for automated detection and classification of plant leaf diseases using classical image processing and machine learning methods. The pipeline included image acquisition, preprocessing (grayscale conversion, noise filtering), segmentation (Otsu thresholding), and feature extraction using texture (GLCM) and color-based descriptors. Trained a Support Vector Machine (SVM) classifier to distinguish between multiple disease categories with high accuracy. This project demonstrated practical application of computer vision and pattern recognition in agriculture, contributing to early disease detection and low-cost crop health monitoring solutions for precision farming.